

Sycophancy Under Deliberative Scaffolding: Weak Propositional Probes, Reliable Social Metrics, and What Moral Sycophancy Actually Measures

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Abstract

Frontier language models are widely criticised for sycophancy, yet benchmark choice matters. Hand-written Sharma probes saturate at $\leq 2\%$ on our quartet; on BrokenMath (Petrov et al., 2025) the same models show 39–83% sycophancy ($n=451$), confirming the propositional floor is a weak-instrument artifact. We evaluate Narration-of-Thought (NoT) against raw, IO, standard CoT, and multi-stakeholder NoT on ELEPHANT (Cheng et al., 2025) ($n=150$ per slice, seed 44). NoT cuts OEQ validation on all four generators (Fisher $p < 10^{-4}$; -26 to -53 percentage points vs. CoT), landing within two points of the crowdsourced human rate on three of four. **The same scaffold does not transfer to propositional sycophancy:** on BrokenMath, NoT moves the Sycophant rate significantly on only one of three generators (grok, -12 pp, $p < 0.001$; haiku/sonnet null). **Moral both-NTA does not replicate earlier collapse claims:** a loader bug had fed empty flip-pair prompts; corrected runs show null-to-mixed both-NTA with no NoT reduction under binary extraction, and Phase 17’s free-form arm reveals the opposite sign—NoT *raises* both-NTA ($34\% \rightarrow 67\%$ pooled), confirming a distinct cross-prompt construct. Phase 14 narrative gradients targeting social sycophancy reduce holdout validation from $76\% \rightarrow 0.7\%$ (training judge), beating TextGrad, OPRO, and APE baselines; framing is corroborated cross-judge while indirectness is judge-dependent (5% vs. 81%). A three-family judge panel with Krippendorff’s α (Krippendorff, 2004) flags all three social metrics as low inter-judge reliability ($\alpha \leq 0.42$), with validation alone showing strong judge-vs-human agreement (κ 0.73 – 0.86). We conclude that deliberative scaffolding targets *social* single-response face-preservation; propositional and moral-cross-prompt metrics require stronger instruments and construct clarity.

1 Introduction

Sycophancy in language models—agreeing with users even when they are wrong—is a recognised alignment failure (Sharma et al., 2024; Malmqvist, 2024). Mitigation efforts often assume a single construct, yet recent work distinguishes *propositional* sycophancy (endorsing false factual claims) from *social* sycophancy (preserving the user’s face through validation, indirectness, framing, or moral affirmation) (Cheng et al., 2025). Our parent work introduces Narration-of-Thought (NoT), a system prompt that forces stakeholder enumeration, consequence projection, explicit uncertainty, and a committed decision before answering ethical dilemmas.

An early Sharma replication on our quartet showed near-zero propositional sycophancy, suggesting instrument saturation. Phases 15–16 deepen this finding: BrokenMath’s expert-verified false-premise olympiad problems (Petrov et al., 2025) and SycEval’s graded probes (Fanous et al., 2025) reveal substantial headroom on frontier models that obvious-falsehood probes miss. Phases 15 and 17 address two measurement crises in the social benchmark: judge-dependent indirectness scores and a confounded moral arm that suppressed the NoT scaffold via a binary verdict suffix.

This paper makes four contributions. (1) We document Sharma probe saturation and adopt BrokenMath as the rigorous propositional complement. (2) We report scaled ELEPHANT results with corrected moral flip-pair loading and a free-form moral arm. (3) We run narrative-gradient optimisation on social sycophancy (Phase 14) and a cross-family judge reliability panel (Phase 15). (4) We separate constructs: validation/framing are single-response metrics; moral both-NTA is cross-prompt consistency and behaves differently under binary vs. free-form extraction.

2 Background

2.1 Propositional sycophancy

Sharma et al. (Sharma et al., 2024) elicit agreement with false premises via opinion-mirroring and false-premise probes. BrokenMath (Petrov et al., 2025) extends this to 451 expert-verified adversarial theorem-proving problems with a four-category judge (Sycophant / Detected / Corrected / Ideal). SycEval (Fanous et al., 2025) reports 29–58% sycophancy on plausible false premises across frontier models.

2.2 Social sycophancy

ELEPHANT (Cheng et al., 2025) scores validation, indirectness, framing, and moral both-NTA on advice. Consumer LLMs exceed human baselines by roughly 47 percentage points on OEQ validation; our question is whether a deliberative scaffold closes that gap.

2.3 Narration-of-Thought

NoT requires five narrative sections before the final answer. Multi-stakeholder NoT runs the same agent prompt through a five-round integration protocol with a cross-vendor moderator.

3 Method

3.1 Generators and arms

Generators (quartet): gpt-5.4-nano, claude-haiku-4-5, claude-sonnet-4-6, grok-4-1-fast-reasoning. **Single-agent arms:** raw (empty system prompt), IO, standard CoT, NoT (narrative_cot), plus depth-optimised NoT-v2/v3 (Phase 14). **Moral modes (Phase 17):** *binary* appends Output only YTA or NTA (ELEPHANT default); *free_form* allows full reasoning then LLM verdict extraction. **Multi-stakeholder arm:** NoT agents + claude-sonnet-4-6 moderator.

3.2 Data

OSF ELEPHANT full release; stratified $n=150$ per slice (seed 44). BrokenMath benchmark split ($n=451$ adversarial) via HuggingFace INSAIT-Institute/BrokenMath.

3.3 Scoring

Faithful ELEPHANT judge prompts in elephant_scorers.py; default judge claude-haiku-4-5. BrokenMath four-category

Generator	Std CoT	NoT
gpt-5.4-nano	0.0%	2.2%
claude-haiku-4-5	1.1%	0.0%
claude-sonnet-4-6	0.0%	0.0%
grok-4-1-fast-reasoning	0.0%	0.0%

Table 1: Sharma SycophancyEval sycophancy rate ($N=90$ probes per cell). Saturated floor; not used for scaffold claims.

judge in brokenmath_scorer.py. Phase 15 panel: gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning on $n=60$ stratified OEQ responses; Krippendorff’s α and Cohen’s κ vs. crowdsourced human gold (data/judge_gold.jsonl).

4 Results

4.1 Sharma replication: obvious-falsehood floor

Table 1 summarises SycophancyEval (30 obvious-falsehood probes). Every cell is $\leq 2.2\%$; the instrument cannot discriminate arms on propositional agreement.

4.2 BrokenMath: graded propositional headroom (Phase 16)

We evaluate quartet $\times$ {raw, standard CoT, NoT} on BrokenMath’s 451 adversarial problems with a four-category judge (Sycophant / Detected / Corrected / Ideal). Unlike Sharma, BrokenMath is designed for plausible false premises in competition-level mathematics.

The instrument is not saturated. Standard-CoT Sycophant rates span 51% (grok) to 83% (haiku)—two orders of magnitude above the Sharma floor (Table 2). **NoT does not generalise to propositional sycophancy.** The validation/framing reductions that hold on ELEPHANT do *not* transfer: NoT moves the Sycophant rate by -3.5 pp on haiku ($p=0.20$) and $+3.1$ pp on sonnet ($p=0.34$), neither significant. The single exception is grok, where NoT cuts sycophancy $51\% \rightarrow 39\%$ (-12 pp, Fisher $p<0.001$)—the only generator that exposes substantial *Detected* and *Ideal* headroom under the scaffold. This sharpens the paper’s central claim: deliberative scaffolding targets social face-preservation, not the propositional error-detection that BrokenMath measures.

Generator	Raw	Std CoT	NoT	$\Delta(\text{NoT}-\text{CoT})$
haiku	81%	83%	80%	-3.5
sonnet	70%	69%	73%	+3.1
grok	48%	51%	39%	-12.0*

Table 2: BrokenMath Sycophant rate ($n=451$ adversarial problems, judge claude-haiku-4-5). Rates far exceed the Sharma floor; NoT reduces sycophancy significantly only on grok (*Fisher $p<0.001$). gpt-5.4-nano is excluded: as a reasoning model it requires a large generation budget on competition mathematics that is impractical at our scale, and is deferred to an extended run.

4.3 ELEPHANT single-agent: validation and framing

Validation: the headline effect. NoT cuts OEQ validation on all four generators (Fisher $p < 10^{-4}$; -26 to -53 pp vs. CoT), meeting the pre-registered falsification criterion. Three generators land within two points of the 29% human rate; sonnet reaches 47%.

Framing. NoT lowers framing on every generator (significant on three of four on OEQ/AITA), and on SS premise-acceptance (-13 to -24 pp vs. CoT).

Moral sycophancy: retraction and correction. An implementation error in `load_elephant.py` read a nonexistent prompt column for flip pairs, yielding `prompt_len=0` and models seeing only the binary suffix. **All prior moral-collapse claims are retracted.** With corrected loading (`original_post / flipped_story`), both-NTA rates are null-to-mixed: e.g. nano $\approx 50\%$ across raw/CoT/NoT; haiku $\approx 15\text{--}23\%$. NoT does not systematically reduce moral both-NTA under binary extraction (Table 5, corrected).

Phase 17 free-form moral arm. Removing AITA_BINARY_SUFFIX lets NoT operate on full stories; verdicts are extracted post hoc by an LLM. The full quartet run ($n=150$ pairs per cell) shows that free-form both-NTA is *higher* than binary mode and that **NoT increases rather than decreases moral both-NTA** once the scaffold is un-suppressed (Table 3). Pooled both-NTA rises from 34% (binary) to 67% (free-form); under NoT, haiku moves 15% $\rightarrow$ 93% and sonnet 17% $\rightarrow$ 72%. This is the opposite sign of the validation/framing effect and confirms moral both-NTA measures cross-prompt consistency rather than single-response face-preservation: when NoT

Generator	Raw	Std CoT	NoT
gpt-5.4-nano	78%	79%	78%
claude-haiku-4-5	59%	53%	93%
claude-sonnet-4-6	32%	38%	72%
grok-4-1-fast-reasoning	70%	73%	82%

Table 3: Phase 17 free-form moral both-NTA rate ($n=150$ pairs). NoT raises both-NTA on every generator (sharply on haiku/sonnet), the opposite of its validation/framing effect.

Metric (slice)	Human	Raw	IO	Std CoT	NoT
OEQ validation	29%	66%	54%	63%	27%
OEQ framing	61%	55%	65%	65%	23%
AITA validation	7%	47%	43%	37%	17%
AITA framing	41%	50%	52%	52%	22%
Moral both-NTA	-	18%	25%	23%	15%

Table 4: ELEPHANT rates on $n=150$ OSF subsample, claude-haiku-4-5 (corrected flip-pair loading). Moral row: binary both-NTA; no collapse under NoT.

enumerates stakeholders and refuses to take sides, it affirms *both* parties of a flipped conflict more often, not less.

Indirectness. NoT nudges indirectness up on several cells; multi-stakeholder debate recovers indirectness (Section 4.4).

4.4 Multi-stakeholder NoT recovers the indirectness cost

Debate-scored integrated advice lowers indirectness in all eight dataset-generator cells (-11 to -28 pp) while preserving validation gains.

5 Phase 14: Narrative Gradients for Social Sycophancy

We optimise NoT with textual gradients on $L = \text{mean}(\text{validation} + \text{indirectness} + \text{framing})$ over a disjoint OEQ train split ($n=100$; indices 150–249). Holdout: Phase 13 OEQ ($n=150$) with claude-haiku-4-5 generator.

On holdout, hand NoT shows validation 76%, framing 39%, and indirectness 16% under the training judge (grok-4-1-fast-reasoning; pooled loss 0.44). The optimised prompt reduces validation to 0.7%, framing to 3%, and indirectness to 5% (training judge; $n=148$; loss 0.03). A third judge (claude-haiku-4-5) corroborates validation (2%) and framing (5%) but scores indirectness at 81% vs. 5% under the training judge—the same contested pattern that motivates Phase 15.

	nano	haiku	sonnet	grok
<i>OEQ validation (human =29%)</i>				
Raw	78%	66%	77%	83%
Std CoT	75%	63%	73%	82%
NoT	27%	27%	47%	29%
<i>Moral both-NTA (binary; corrected loader)</i>				
Raw	51%	18%	25%	31%
Std CoT	50%	23%	24%	32%
NoT	51%	15%	17%	42%

Table 5: Cross-vendor replication. Top: OEQ validation (NoT reductions significant on all four). Bottom: corrected moral both-NTA—no systematic NoT effect; construct differs from validation/framing.

Optimiser	Init	Holdout loss
Narrative gradients	NoT	0.029
APE	NoT	0.104
TextGrad-CoT	Std CoT	0.170
OPRO	NoT	0.200
Hand NoT (reference)	—	0.436

Table 6: Phase 14 optimiser holdout loss (validation + indirectness + framing, training judge grok-4-1-fast-reasoning, $n \approx 148$ –150). Lower is better. Narrative gradients win; all four optimisers improve substantially on hand NoT.

Optimiser comparison. We benchmark narrative gradients against three matched-budget baselines (~ 10 prompt evaluations each): TextGrad on standard CoT, OPRO, and APE (Table 6). Narrative gradients achieve the lowest holdout loss (0.03), beating APE (0.10), TextGrad-CoT (0.17), and OPRO (0.20). This supports the pre-registered hypothesis that scaffold structure—not merely iterative optimisation—carries the social-sycophancy reduction: the two narrative-initialised methods (narrative grad, APE) outperform the CoT-initialised TextGrad baseline, and trajectory-based narrative gradients beat the others outright.

The gradient rewrites the protagonist section into a “Premise Audit” that challenges embedded assumptions—a mechanistic diagnosis of why first-person NoT validation fails. Degeneracy checks show stakeholder count and uncertainty scores remain near hand NoT.

6 Phase 15: Judge Reliability Panel

We re-score $n=60$ stratified OEQ responses with three judge families (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning) on validation, indirectness, and framing. Reporting uses Krippen-

Metric	Inter-judge α ($n=60$)	κ vs. human		
		nano	haiku	grok
Validation	0.42	0.85	0.86	0.73
Indirectness	−0.23	0.31	0.39	0.63
Framing	0.11	0.15	0.57	0.70

Table 7: Phase 15 judge reliability. Inter-judge Krippendorff α over the three-family panel; Cohen’s κ of each judge against crowdsourced human labels ($n=30$ human responses). No metric clears $\alpha=0.67$; validation alone shows strong judge-vs-human agreement.

dorff’s α ; metrics below $\alpha=0.67$ require majority vote or are flagged low-reliability. Crowdsourced human OEQ labels (judge_gold.jsonl, $n=30$) provide Cohen’s κ against each judge, scored on the *human* responses directly.

Inter-judge reliability is low on all three metrics (Table 7): validation $\alpha=0.42$, framing $\alpha=0.11$, indirectness $\alpha=-0.23$. None clears the 0.67 threshold, so panel headline rates use majority vote. The picture is more favourable against human gold: validation agreement is strong for every judge (Cohen’s κ vs. human 0.73–0.86), indirectness is moderate and judge-dependent (0.31–0.63), and framing is mixed (0.15–0.70). The contrast—high judge-vs-human κ on validation but only moderate *inter-judge* α —reflects the narrow positive base rate among the gold human responses; the practical implication is that validation is the only ELEPHANT metric we report without a reliability caveat, and indirectness reductions from optimised prompts (Section 5) are not reportable from any single judge.

7 Discussion

Dual benchmarks, correctly framed. Sharma saturates at $\leq 2\%$; BrokenMath does not (39–83%). Crucially, the scaffold that reliably reduces *social* sycophancy on ELEPHANT does not transfer to BrokenMath’s *propositional* error-detection (significant only on grok), evidence that the two are distinct constructs rather than a single “sycophancy” axis. Social ELEPHANT metrics discriminate NoT on validation and framing with strong cross-vendor evidence.

Moral sycophancy is a different construct. Both-NTA measures whether a model affirms *both* sides of a flipped conflict across prompts—not whether a single response validates the user. Binary suffix mode further confounds the scaffold. Researchers should not pool moral rates with vali-

dation/framing claims.

Judge reliability matters. Indirectness reductions from optimised prompts are not reportable without a panel or human gold; validation survives cross-judge checks against human labels, while framing is mixed.

8 Conclusion

NoT reduces social validation and framing sycophancy across four generators where Sharma probes show no signal. Moral both-NTA does not follow the same pattern once loading and construct issues are corrected. BrokenMath supplies the propositional headroom Sharma lacks; a judge panel supplies the reliability gate indirectness requires. Together, these phases narrow the defensible claim: deliberative scaffolding targets single-response social face-preservation, not propositional agreement or cross-prompt moral consistency.

Limitations

All sycophancy outcomes are scored by LLM judges; while Phase 15 validates the validation metric against human gold (κ 0.73–0.86), indirectness and framing carry low inter-judge reliability and should be read as judge-conditioned. The optimiser comparison uses a single training judge (grok-4-1-fast-reasoning) and a single generator (claude-haiku-4-5) for tractability; cross-generator optimisation is left to future work. The Phase 14 result that narrative gradients beat other optimisers isolates *outcome*, not *mechanism*: we have not yet ablated which narrative sections carry the effect, so causal attribution to specific scaffold components remains open. BrokenMath uses a single judge family, and the free-form moral arm adds a second verdict-extraction call whose own reliability we do not separately audit. Sample sizes are $n=150$ per ELEPHANT slice and $n=451$ for BrokenMath; effects below a few points may be underpowered. All generators are accessed through a single hosted provider, and results may not transfer to other deployments or model versions.

Ethical Considerations

This work studies sycophancy mitigation and does not involve new human-subjects data collection; human labels are drawn from the public ELEPHANT/OSF release and used only as reliability

gold. Reducing sycophancy is dual-use: a scaffold that makes models less validating can also make them blunt or dismissive in genuinely sensitive contexts, and the Phase 17 finding that NoT *increases* moral both-NTA shows that “less sycophantic” is not uniformly desirable. We therefore frame NoT as a measurement and construct-clarification tool, not a deployment-ready safety intervention, and we caution against optimising single sycophancy metrics in isolation. No personally identifying information is processed, and all adversarial math problems come from the public BrokenMath benchmark.

Reproducibility

Phases 13–17 pre-registration:
Guidance_Documents/study_design.md.
Scripts: run_elephant.py
(-moral-mode), judge_panel.py,
run_brokenmath.py, run_phase14_*.py.
Outputs: elephant_summary.json,
brokenmath_summary.json,
judge_reliability_summary.json,
phase14_summary.json.

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